WEAVE

CENTRE FOR INTEGRATIVE PSYCHIATRY

WEAVE PSYCHOTHERAPY — VOL. 5

Behaviour Therapy

Conditioning, Exposure, and Behavioral Activation

Classical Conditioning · Operant Conditioning · Exposure · Desensitization · Behavioral Activation · Social Learning

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Deep Study

Behaviour Therapy — Weave Psychotherapy Vol. 5

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1. HISTORICAL DEVELOPMENT

1.1 From Laboratory to Clinic

Behaviour therapy emerged from a deliberate rejection of psychoanalytic introspection in favour of observable, measurable behaviour change. Its roots are entirely empirical — each major figure brought laboratory science into the consulting room.

Ivan Pavlov (1849–1936) established the foundation. Working in his St. Petersburg laboratory on digestive physiology (for which he won the 1904 Nobel Prize), he observed that dogs began salivating not just to food but to stimuli

associated with food — the laboratory assistant's footsteps, the sound of a metronome. He called this a "conditioned reflex" and spent decades systematically mapping the laws of acquisition, extinction, spontaneous recovery, stimulus generalisation, and discrimination. Pavlov never practised psychotherapy, but his work gave the field its first mechanistic language for learned responses.

John B. Watson (1878–1958) translated Pavlovian conditioning into a manifesto. His 1913 paper "Psychology as the Behaviorist Views It" declared that psychology must abandon introspection and study only observable behaviour. In 1920, Watson and Rosalie Rayner demonstrated conditioned fear in "Little Albert" — an 11-month-old infant who was taught to fear a white rat by pairing it with a loud noise. The experiment was ethically indefensible by modern standards, but it proved that emotional responses could be acquired through conditioning.

> EXAM PEARL
>
> Watson's Little Albert experiment demonstrated fear acquisition through classical conditioning. Mary Cover Jones (1924) demonstrated fear elimination in "Little Peter" using counterconditioning — pairing a feared rabbit with food (the child's favourite). Jones is often called the "mother of behaviour therapy."

B.F. Skinner (1904–1990) shifted the focus from respondent (Pavlovian) conditioning to operant conditioning. His key insight: behaviour is controlled primarily by its consequences, not by preceding stimuli. Through decades of work with pigeons and rats in the "Skinner box," he mapped the laws of reinforcement, punishment, shaping, extinction, and schedules of reinforcement. His 1953 book *Science and Human Behavior* extended these principles to human psychology. Skinner never used the word "therapy," preferring "behaviour modification."

Joseph Wolpe (1915–1997) was the first to build a complete clinical method from conditioning principles. A South African psychiatrist, Wolpe demonstrated in the 1950s that neurotic anxiety in cats could be eliminated by feeding them in

progressively more anxiety-provoking environments — the principle of reciprocal inhibition. His 1958 book *Psychotherapy by Reciprocal Inhibition* introduced systematic desensitization and launched behaviour therapy as a clinical discipline.

Albert Bandura (b. 1925) expanded the behavioural framework beyond direct conditioning. His 1961 Bobo doll experiment showed that children could acquire aggressive behaviours simply by watching a model — no direct reinforcement required. His 1977 *Social Learning Theory* introduced self-efficacy and reciprocal determinism, bridging behaviourism and cognitive psychology.

Isaac Marks (b. 1935) brought exposure therapy into mainstream psychiatry at the Maudsley Hospital, London. His 1987 book *Fears, Phobias, and Rituals* synthesised decades of clinical research demonstrating that direct exposure — not relaxation, not insight — was the active ingredient in fear reduction. Marks' work on exposure and response prevention (ERP) for OCD remains the gold standard.

> EXAM PEARL
>
> The "three waves" of behaviour therapy: (1) Classical and operant conditioning (1950s–1960s), (2) Cognitive-behavioural integration (1970s–1980s, Beck, Ellis), (3) Acceptance and mindfulness-based approaches (1990s–present, ACT, MBCT, DBT). Each wave added to rather than replaced the previous one.

1.2 Timeline of Key Developments

Year	Figure	Contribution
1897	Pavlov	Classical conditioning described
1913	Watson	Behaviourist manifesto
1920	Watson & Rayner	Little Albert experiment
1924	Mary Cover Jones	Little Peter — counterconditioning
1938	Skinner	*The Behavior of Organisms* — operant conditioning
1953	Skinner	*Science and Human Behavior*
1958	Wolpe	*Psychotherapy by Reciprocal Inhibition*
1961	Bandura	Bobo doll experiment
1966	Azrin & Ayllon	Token economy in psychiatric hospitals
1969	Bandura	*Principles of Behavior Modification*
1974	Lewinsohn	Behavioural model of depression
1977	Bandura	Self-efficacy theory
1986	Foa & Kozak	Emotional processing theory of exposure
1987	Marks	*Fears, Phobias, and Rituals*
1996	Craske & Barlow	Inhibitory learning model of exposure
2001	Martell et al.	Behavioural Activation for Depression

2. CLASSICAL CONDITIONING

2.1 Pavlov's Experimental Paradigm

Classical (respondent) conditioning involves learning through association. An initially neutral stimulus acquires the ability to elicit a response by being paired with a stimulus that naturally produces that response.

Key terms:

- **Unconditioned stimulus (US):** Naturally elicits a response without learning (e.g., food → salivation)
- **Unconditioned response (UR):** The natural, unlearned response to the US
- **Conditioned stimulus (CS):** A previously neutral stimulus that, after pairing with the US, comes to elicit a response
- **Conditioned response (CR):** The learned response to the CS

> KEY INSIGHT
>
> **MNEMONIC — "UNCLES CONDITION NEPHEWS":** UNConditioned = natural/unlearned. Conditioned = learned. The US is the uncle (naturally powerful), the CS is the nephew (acquires power through association).

2.2 Core Phenomena

Acquisition — the phase during which the CS-US pairing is established. Optimal conditioning occurs when the CS precedes the US by approximately 0.5 seconds (forward delay conditioning). Simultaneous presentation produces weaker conditioning; backward pairing (US before CS) typically produces no conditioning.

Extinction — repeated presentation of the CS without the US leads to gradual weakening of the CR. Critically, extinction is NOT "unlearning." The original CS-US association remains intact; extinction creates a new, inhibitory association

(CS → no US) that competes with the original. This has profound clinical implications for exposure therapy.

> EXAM PEARL
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> Extinction is new learning, not erasure. The original fear memory is not deleted — it is suppressed by a new inhibitory memory. This explains why extinguished fears can return (spontaneous recovery, renewal, reinstatement). Clinically, this is why relapse occurs after successful exposure therapy and why exposure must be conducted in multiple contexts.

Spontaneous recovery — after a rest period following extinction, the CR reappears at reduced strength. This demonstrates that the original association survives extinction.

Stimulus generalisation — the CR is elicited not only by the original CS but by stimuli similar to it. The more similar the stimulus, the stronger the response (generalisation gradient). Clinically, this explains why a person bitten by a German Shepherd may fear all dogs, or why a trauma survivor may be triggered by any loud noise, not just gunshots.

Stimulus discrimination — the organism learns to respond to the CS but not to similar stimuli. Achieved through differential reinforcement (CS+ paired with US; CS- not paired). Pavlov demonstrated that forcing extremely fine discriminations produced "experimental neurosis" in dogs — agitation, aggression, and breakdown of previously established conditioned responses.

> CLINICAL ANCHOR
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> Pavlov's experimental neurosis is a laboratory analogue of anxiety disorders. When the environment becomes unpredictable — when the organism cannot discriminate safe from dangerous — the result is chronic autonomic arousal and behavioural disorganisation. This maps directly onto the experience of generalised anxiety disorder, where the patient cannot discriminate genuine threats from safe situations.

Higher-order conditioning — once a CS reliably elicits a CR, it can serve as a US to condition a new, second-order CS. A metronome (CS1) paired with food produces salivation. A light (CS2) paired with the metronome then also produces salivation — without ever being directly paired with food. Higher-order conditioning is weaker and extinguishes faster, but it explains how complex fear networks develop in anxiety disorders.

2.3 Clinical Applications of Classical Conditioning

Phenomenon	Clinical Application
Acquisition	Fear conditioning in phobias, PTSD
Extinction	Exposure therapy, systematic desensitization
Spontaneous recovery	Relapse after successful treatment
Generalisation	Spread of phobic avoidance, trigger expansion in PTSD
Discrimination	Discriminative exposure (safety vs. danger signals)
Higher-order conditioning	Complex fear networks, conditioned drug cues
Interoceptive conditioning	Panic disorder (bodily sensations as CS for fear)

EXAM STRATEGY

When answering questions on classical conditioning in clinical contexts, always distinguish between the acquisition model (how symptoms develop) and the extinction model (how treatment works). Examiners test whether you understand that exposure therapy works through extinction — and that extinction is new learning, not erasure.

3. OPERANT CONDITIONING

3.1 Skinner's Framework

Operant conditioning governs behaviour controlled by its consequences. Unlike classical conditioning (where the organism is passive), operant conditioning involves the organism acting on its environment and being shaped by the results.

The Four Contingencies

Stimulus Added	Stimulus Removed	
Behaviour Increases	Positive Reinforcement (add something desirable)	Negative Reinforcement (remove something aversive)
Behaviour Decreases	Positive Punishment (add something aversive)	Negative Punishment (remove something desirable)

KEY INSIGHT

MNEMONIC — "POSITIVE = PLUS, NEGATIVE = MINUS": "Positive" and "negative" do NOT mean good and bad. Positive = adding a stimulus. Negative = removing a stimulus. "Reinforcement" = behaviour increases. "Punishment" = behaviour decreases. This 2×2 grid is the entire system.

EXAM PEARL

The most commonly confused pair: negative reinforcement vs. punishment. Negative reinforcement INCREASES behaviour by removing an aversive stimulus. A patient takes a benzodiazepine (behaviour) → anxiety decreases (aversive stimulus removed) → pill-taking increases. This is negative reinforcement, not positive reinforcement (nothing pleasant was added) and not punishment (the behaviour increased, not decreased).

3.2 Schedules of Reinforcement

The pattern of reinforcement delivery profoundly affects behaviour.

Schedule	Definition	Response Pattern	Extinction Rate	Clinical Example
Continuous (CRF)	Every response reinforced	Steady but moderate rate	Fastest	Teaching new behaviours — every correct response praised
Fixed Ratio (FR)	Reinforcement after set number of responses	High rate with post-reinforcement pause	Fast	Piecework pay; token economy (e.g., 5 tasks = 1 token)
Variable Ratio (VR)	Reinforcement after unpredictable number of responses	Highest, steadiest rate	Slowest (most resistant)	Gambling; intermittent reassurance-seeking in OCD
Fixed Interval (FI)	Reinforcement after set time period	"Scallop" — accelerating near reinforcement time	Moderate	Checking clock before appointment; medication schedules
Variable Interval (VI)	Reinforcement after unpredictable time period	Steady, moderate rate	Slow	Random supervisor visits; slot machine timing element

EXAM PEARL

Variable ratio schedules produce the most persistent behaviour and the slowest extinction. This is why gambling is so addictive (reinforcement is unpredictable in timing and magnitude) and why intermittent partner abuse creates powerful trauma bonds (affection is delivered on a VR schedule).

> CLINICAL ANCHOR
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> In anxiety disorders, avoidance behaviour is maintained by negative reinforcement on a continuous schedule — every avoidance reduces anxiety. Safety behaviours in social anxiety, reassurance-seeking in OCD, and escape in panic disorder all follow this pattern. The clinical task is to break this reinforcement contingency through exposure (preventing the avoidance and allowing anxiety to habituate).

3.3 Shaping and Chaining

Shaping (successive approximations) — reinforcing behaviours that progressively resemble the target behaviour. Used when the target behaviour never occurs spontaneously and therefore cannot be directly reinforced. Wolberg notes: "Start with response remotely similar to target, differentially reinforce as it approaches the goal."

Chaining — linking a sequence of individual behaviours into a complex chain where each step serves as both a reinforcer for the previous step and a discriminative stimulus for the next. Taught using backward chaining (last step first) or forward chaining (first step first). Used in rehabilitation, developmental disabilities, and skills training.

3.4 Clinical Applications

- **Token economy** — systematic application of operant principles in institutional settings (psychiatric wards, classrooms, prisons)
- **Contingency management** — reinforcement of drug-free urines in substance use treatment
- **Behavioural parent training** — teaching parents to use differential reinforcement instead of punishment
- **Applied behaviour analysis (ABA)** — structured operant conditioning for autism spectrum disorder
- **Habit reversal training** — for tics, trichotillomania, skin-picking

4. SOCIAL LEARNING THEORY

4.1 Bandura's Contribution

Albert Bandura's social learning theory (later renamed social cognitive theory) was a direct challenge to radical behaviourism's insistence that all learning requires direct reinforcement. Bandura demonstrated that humans learn extensively through observation — watching what others do and what happens to them.

The Bobo Doll Experiment (1961)

Children watched an adult model either (a) attack an inflatable Bobo doll (punching, kicking, hitting with a mallet while shouting "Sock him!"), (b) play quietly with other toys, or (c) no model exposure. When subsequently placed in a room with the Bobo doll, children who had watched the aggressive model reproduced specific aggressive acts — including novel acts the model had performed. No direct reinforcement was given to the children at any point.

> EXAM PEARL
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> The Bobo doll experiment demonstrated observational learning (vicarious conditioning). A follow-up study showed that consequences to the MODEL mattered: children who saw the model rewarded for aggression imitated more; children who saw the model punished imitated less. But when offered incentives, all groups could reproduce the behaviours equally — proving that learning and performance are distinct. Children learned regardless of consequences; they performed based on expected consequences.

4.2 Key Concepts

Observational learning (modelling) — requires four processes:

1. **Attention** — observer must attend to the model's behaviour
2. **Retention** — observer must encode and remember the behaviour
3. **Reproduction** — observer must be capable of performing the behaviour

4. **Motivation** — observer must have reason to perform the behaviour

Self-efficacy — the individual's belief in their capacity to execute the behaviours needed to produce a specific outcome. Self-efficacy is domain-specific (a person may have high self-efficacy for driving but low for public speaking). It is the single strongest predictor of whether someone will attempt a feared task — more predictive than actual skill level.

> CLINICAL ANCHOR
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> Self-efficacy is a critical mediator of exposure therapy outcomes. A patient who completes an exposure hierarchy successfully but attributes the success to luck or the therapist's presence will not develop self-efficacy and will relapse. The therapist must explicitly process attributions: "You did that. Not me, not the medication — you walked into that situation and stayed."

Reciprocal determinism — behaviour, personal factors (cognitions, emotions, biology), and the environment all influence each other bidirectionally. This was Bandura's fundamental critique of both radical behaviourism (environment → behaviour) and psychodynamic theory (internal drives → behaviour). All three components are in constant, reciprocal interaction.

Sources of self-efficacy (in order of potency):

1. **Mastery experiences** — personal success (strongest source)
2. **Vicarious experiences** — seeing others succeed
3. **Verbal persuasion** — encouragement from others
4. **Physiological/emotional states** — interpreting arousal as capability or incapability

4.3 Clinical Applications of Modelling

- **Participant modelling** — therapist demonstrates approach to feared stimulus, then guides patient through graded exposure (Bandura, 1969). More effective than observation alone.

- **Coping models** — models who display initial anxiety but gradually master the task are more effective than "mastery models" who show no distress. The coping model normalises anxiety and demonstrates that it is manageable.
- **Social skills training** — modelling, role-play, and feedback to teach interpersonal behaviours. Used in schizophrenia, social anxiety, autism spectrum disorder.
- **Video self-modelling** — patients watch edited videos of themselves performing desired behaviours successfully.

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5. SYSTEMATIC DESENSITIZATION

5.1 Wolpe's Reciprocal Inhibition

Joseph Wolpe's systematic desensitization is built on the principle of reciprocal inhibition: "If a response inhibitory of anxiety can be made to occur in the presence of anxiety-evoking stimuli, it will weaken the bond between these stimuli and the anxiety" (Wolpe, 1958).

The logic: anxiety and relaxation are physiologically incompatible states (opposing autonomic activation). If you can teach a patient to be deeply relaxed while imagining a feared stimulus, the relaxation response will inhibit the anxiety response and weaken the conditioned fear.

5.2 Three Steps

Step	Procedure	Details
1. Relaxation training	Teach progressive muscle relaxation (PMR)	16 muscle groups → 7 → 4; typically 4–6 sessions to master
2. Anxiety hierarchy	Construct a ranked list of feared situations	10–20 items, rated 0–100 SUDs; evenly spaced across the range
3. Graded imaginal exposure	Pair relaxation with imagined feared scenes	Start at lowest item; progress when anxiety = 0; 3–4 scenes per session

EXAM PEARL

In systematic desensitization, progression up the hierarchy is ALWAYS controlled by the patient's reported anxiety. You never move to the next item until the current item elicits zero anxiety on two consecutive presentations. If anxiety rises above 25 SUDs, the patient signals (e.g., raises a finger) and returns to relaxation before re-attempting the item.

5.3 In-Vivo vs. Imaginal

Imaginal desensitization — patient imagines feared situations while relaxed. Advantages: safe, controllable, practical for situations difficult to recreate (flying, storms). Disadvantage: some patients cannot generate vivid imagery; generalisation to real life may be limited.

In-vivo desensitization — patient confronts real feared stimuli in graded steps while using relaxation skills. Generally more effective than imaginal. Wolpe acknowledged this but used imaginal desensitization when in-vivo was impractical.

> CLINICAL ANCHOR
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> Lazarus (1961) compared systematic desensitization to insight-oriented therapy for phobias: 13/18 patients (72%) recovered with desensitization vs. 2/17 (12%) with insight therapy. This was one of the earliest controlled demonstrations that behaviour therapy outperformed traditional psychotherapy for specific conditions.

5.4 Mechanism Debate

Wolpe attributed the effect to reciprocal inhibition — a Pavlovian counterconditioning process. Later research questioned this: relaxation may not be essential (exposure alone works), and the effect may be mediated by habituation or inhibitory learning rather than counterconditioning. Systematic desensitization has been largely superseded by direct exposure therapy for most anxiety disorders, but it remains useful when patients are too anxious to tolerate direct exposure or when a gradual approach is needed.

6. EXPOSURE THERAPY

6.1 The Central Principle

Exposure therapy is the single most effective psychological intervention for anxiety disorders. The principle is straightforward: systematic, repeated confrontation with feared stimuli in the absence of the feared outcome leads to fear reduction.

> EXAM STRATEGY
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> Any exam question on "most effective psychological treatment for anxiety disorders" almost certainly has exposure therapy (or a variant of it) as the answer. ERP for OCD, prolonged exposure for PTSD, in-vivo exposure for specific phobias, interoceptive exposure for panic disorder.

6.2 Theoretical Models

Habituation model — repeated, prolonged exposure leads to a natural decrease in the fear response. Fear peaks and then declines within a session (within-session habituation) and across sessions (between-session habituation). Limitation: habituation does not fully explain fear return or context-dependent relapse.

Emotional processing theory (Foa & Kozak, 1986) — effective exposure requires: (1) activation of the fear structure (the patient must actually feel afraid), (2) incorporation of information incompatible with the pathological elements of the fear structure. Two indicators of successful processing: within-session habituation and between-session habituation.

Inhibitory learning model (Craske et al., 2008) — the dominant current model. Exposure does not erase the original fear memory. Instead, it creates a new, inhibitory association (CS → no US) that competes with the original excitatory association (CS → US). The original fear memory and the new safety memory coexist; the context determines which is retrieved.

> EXAM PEARL
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> The inhibitory learning model has changed how we deliver exposure therapy. Key clinical implications: (1) maximise expectancy violation (do not use relaxation during exposure — let anxiety rise), (2) vary contexts (do exposure in multiple locations, times, states), (3) use deepened extinction (combine multiple feared stimuli), (4) occasional reinforced extinction (sometimes pair CS with US to prevent prediction-based safety learning), (5) remove safety signals (no therapist reassurance, no subtle avoidance).

6.3 Types of Exposure

Type	Method	Primary Indication
In-vivo	Direct confrontation with real feared stimulus	Specific phobias, social anxiety, agoraphobia
Imaginal	Vivid mental imagery of feared scenario	PTSD (prolonged exposure), fears that cannot be recreated
Interoceptive	Deliberate induction of feared body sensations	Panic disorder (e.g., hyperventilation, spinning, straw-breathing)
Virtual reality (VR)	Computer-simulated feared environment	Phobias (flying, heights), PTSD, where in-vivo is impractical
Flooding	Prolonged, intense exposure to maximum fear stimulus	Rapid fear reduction; high dropout risk
Graded exposure	Systematic progression up a fear hierarchy	Standard approach for most anxiety disorders
Exposure and response prevention (ERP)	Exposure to obsessional trigger + prevention of compulsive ritual	OCD (gold standard treatment)

6.4 Key Procedural Principles

1. **Exposure must be prolonged** — sessions of 45–90 minutes; premature termination reinforces avoidance
2. **Exposure must be repeated** — multiple sessions (typically 8–20) needed for between-session habituation
3. **Exposure must be predictable and controlled** — patient retains sense of agency
4. **Safety behaviours must be eliminated** — subtle avoidance (e.g., carrying a water bottle "just in case," avoiding eye contact during social exposure)

maintains the fear structure

5. **Expectancy violation is key** — the patient must learn that the feared outcome does NOT happen

> CLINICAL ANCHOR
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> The biggest error therapists make with exposure is ending it too early. If a patient with contamination OCD touches a doorknob and immediately washes their hands, the exposure is not just ineffective — it is counter-therapeutic, because it reinforces the belief that touching the doorknob was genuinely dangerous and that washing "saved" them. Response prevention is not a supplementary add-on; it is the mechanism by which exposure works in OCD.

> KEY INSIGHT
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> **MNEMONIC — "PROVE IT WRONG":** The goal of every exposure is for the patient to Predict what will happen, Remain in the situation until the prediction is tested, Observe what actually happens, Verify the disconfirmation, and Evaluate what was learned. (Craske et al.)

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7. BEHAVIORAL ACTIVATION

7.1 Lewinsohn's Model

Peter Lewinsohn (1974) proposed that depression results from a low rate of response-contingent positive reinforcement. The depressed person either (a) receives few pleasant events, (b) has few events available that are reinforcing, or (c) lacks the skills to obtain reinforcement from the environment. The resulting withdrawal and inactivity further reduce reinforcement opportunities, creating a self-maintaining cycle.

> PATHWAY
>
> Triggering event → Reduced activity → Fewer positive reinforcements → Low mood → Further withdrawal → Fewer reinforcements → Deepening depression

7.2 Activity Scheduling

Lewinsohn's Pleasant Events Schedule identified activities correlated with positive mood. Treatment involved systematically increasing engagement in pleasant activities. Beck's cognitive therapy incorporated activity scheduling as a core behavioural component — rating activities for both mastery (sense of accomplishment) and pleasure (enjoyment) on a 0–10 scale.

> CLINICAL ANCHOR
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> Beck's activity scheduling principle (described in Beck, 2021, Ch. 7): "Inactivity maintains depression through a vicious cycle: depressed mood leads to negative thoughts, which lead to inactivity, which prevents opportunities for mastery/pleasure/connection, which reinforces depression." Activity scheduling breaks this cycle by increasing contact with reinforcement BEFORE the patient feels motivated. The key insight: action precedes motivation, not the other way around.

7.3 Modern Behavioural Activation (BA)

Martell, Addis, and Dimidjian (2001) developed a stand-alone Behavioural Activation treatment that stripped cognitive therapy back to its behavioural roots. Their rationale came from the landmark component analysis by Jacobson et al. (1996), which found that the behavioural activation component of CBT was as effective as the full CBT package — cognitive restructuring added nothing.

Core principles of modern BA:

1. Focus on changing what the patient does, not what they think
2. Identify patterns of avoidance and withdrawal
3. Use activity monitoring and scheduling
4. Target activation in line with values and life goals
5. Treat rumination as a form of avoidance behaviour (not a cognitive distortion)
6. Use graded task assignment for overwhelming goals

EXAM PEARL

The Dimidjian et al. (2006) trial compared BA, cognitive therapy, and antidepressant medication (paroxetine) for major depression. For severely depressed patients, BA was as effective as medication and significantly more effective than cognitive therapy. This was a landmark finding because BA is simpler, cheaper, and easier to disseminate than full CBT.

7.4 BA Techniques

Technique	Description
Activity monitoring	Track all activities hour-by-hour; rate mood 0–10
Activity scheduling	Plan specific activities at specific times
Mastery and pleasure ratings	Rate each activity for accomplishment (M) and enjoyment (P)
Graded task assignment	Break large tasks into small, sequential steps
Values assessment	Identify what matters to the patient across life domains
Behavioural experiments	Test negative predictions about activity outcomes
TRAP/TRAC model	Trigger → Response → Avoidance Pattern (TRAP) replaced by Trigger → Response → Alternative Coping (TRAC)

8. AVERSION THERAPY

8.1 Principles

Aversion therapy pairs an undesirable behaviour or stimulus with an aversive consequence, aiming to create a conditioned aversion. The theoretical basis is classical conditioning: if the CS (e.g., taste of alcohol) is repeatedly paired with a

US (e.g., nausea from emetic agent), the CS should come to elicit a CR (nausea, avoidance).

8.2 Methods

Chemical aversion — the most historically significant form. Emetine or apomorphine administered before alcohol consumption produces violent nausea paired with the taste and smell of alcohol. Antabuse (disulfiram) works differently — it is not classical conditioning but an operant deterrent (patient avoids alcohol because they know it will make them ill).

Electrical aversion — mild electric shock paired with the target behaviour or its cues. Used historically for alcohol dependence, sexual deviations, and smoking. Rapidly delivered and precisely controllable, but patient acceptance is low.

Covert sensitization (Cautela, 1967) — the patient imagines the undesirable behaviour and then imagines an intensely aversive consequence (nausea, social humiliation). No physical aversive stimulus is used. More ethical and practical than chemical or electrical methods, but efficacy evidence is limited.

> EXAM PEARL
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> Aversion therapy has largely fallen out of clinical use for several reasons: (1) poor long-term efficacy — conditioned aversions extinguish rapidly, (2) ethical concerns — especially historical misuse in "conversion therapy" for homosexuality, (3) high dropout rates — patients dislike the procedures, (4) better alternatives exist (motivational interviewing, contingency management, pharmacotherapy). The one area where aversion-based principles retain some role is in disulfiram therapy for alcohol dependence, which is more accurately understood as pharmacological deterrence than classical conditioning.

8.3 Ethical Concerns and Current Status

The history of aversion therapy includes deeply unethical applications — "conversion therapy" using electric shocks or emetics to attempt to change sexual orientation. These practices caused significant harm and are now banned

or condemned by every major professional body. Contemporary behaviour therapy has moved decisively toward positive, reinforcement-based approaches. Aversion therapy is included here for historical completeness and because it appears on examinations.

9. BIOFEEDBACK AND RELAXATION

9.1 Progressive Muscle Relaxation (Jacobson, 1938)

Edmund Jacobson developed PMR based on the observation that mental and muscular tension are linked. The technique involves systematically tensing and then releasing muscle groups, training the patient to discriminate between tension and relaxation and to achieve deep muscular relaxation voluntarily.

Standard protocol: 16 muscle groups (dominant hand/forearm, dominant biceps, non-dominant hand/forearm, non-dominant biceps, forehead, upper cheeks/nose, lower cheeks/jaw, neck/throat, chest/shoulders/upper back, abdomen, dominant thigh, dominant calf, dominant foot, non-dominant thigh, non-dominant calf, non-dominant foot) → 7 groups → 4 groups → relaxation by recall (no tension phase) → cue-controlled relaxation (deep breath + cue word).

9.2 Applied Relaxation (Öst, 1987)

Lars-Göran Öst developed applied relaxation as a coping skill for anxiety. The technique trains the patient to achieve rapid relaxation (in 20–30 seconds) and deploy it in anxiety-provoking real-world situations.

Stages: PMR → Release-only relaxation → Cue-controlled relaxation → Differential relaxation → Rapid relaxation → Application in real situations. Total training: 8–12 sessions.

CLINICAL ANCHOR

Applied relaxation is distinct from simple PMR. PMR is a relaxation exercise done at home. Applied relaxation is a coping skill used in real time — the patient learns to detect early signs of anxiety, deploy rapid relaxation, and re-engage with the situation. Öst showed it was as effective as exposure for panic disorder and generalised anxiety disorder.

9.3 Biofeedback

Biofeedback provides real-time physiological data (muscle tension, skin conductance, heart rate, EEG, temperature) to the patient, who then learns to modify these parameters voluntarily.

Type	Measurement	Primary Applications
EMG biofeedback	Muscle tension	Tension headaches, TMJ, chronic pain, bruxism
Thermal biofeedback	Skin temperature	Raynaud's disease, migraine
EEG biofeedback (neurofeedback)	Brainwave patterns	ADHD (controversial), epilepsy, insomnia
HRV biofeedback	Heart rate variability	Anxiety, depression, PTSD, stress management
GSR/EDA biofeedback	Skin conductance	Anxiety, stress, phobias

EXAM STRATEGY

For exam purposes, know that biofeedback has its strongest evidence base for tension-type headaches, Raynaud's disease, and chronic pain (EMG/thermal biofeedback). Evidence for psychiatric applications (anxiety, depression, ADHD) is growing but not yet at the level of first-line recommendation. Wolberg notes Miller's (1978) finding that biofeedback for hypertension had "limited utility."

10. TOKEN ECONOMY AND CONTINGENCY MANAGEMENT

10.1 Token Economy

Developed by Ayllon and Azrin (1968), the token economy is a systematic application of operant conditioning in institutional settings. Patients earn tokens (secondary reinforcers) for performing desired behaviours and exchange them for backup reinforcers (privileges, items, activities).

Key design principles:

1. Target behaviours clearly defined and observable
2. Tokens delivered immediately after desired behaviour
3. Token-to-backup-reinforcer exchange rate is clear and consistent
4. Reinforcers are individualised (what one patient values, another may not)
5. Shaping used for complex behaviours — start by reinforcing approximations

Applications: Originally developed for chronic psychiatric inpatients (schizophrenia), token economies have been used in classrooms, residential facilities, and juvenile justice settings.

> EXAM PEARL
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> The token economy at Anna State Hospital (Ayllon & Azrin, 1968) demonstrated that chronic psychiatric patients who had been considered "unmotivated" or "deteriorated" showed dramatic improvements in self-care, social interaction, and ward behaviour when appropriate reinforcement contingencies were implemented. The implication: what looked like negative symptoms of schizophrenia was partly an artefact of a non-reinforcing institutional environment.

10.2 Contingency Management (CM) in Substance Use

Contingency management is the most effective behavioural intervention for substance use disorders, particularly stimulant use disorders for which no effective pharmacotherapy exists.

Voucher-based reinforcement (Higgins et al., 1991) — patients provide drug-free urine samples and receive vouchers exchangeable for goods/services. Voucher values increase with consecutive negative samples and reset to the initial value if a positive sample occurs (escalating with reset).

> EXAM PEARL
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> Meta-analyses consistently show that contingency management has the largest effect size of any psychosocial intervention for substance use disorders (effect sizes 0.4–0.6). Despite this evidence, implementation remains limited due to: (1) ethical objections to "paying patients not to use drugs," (2) cost concerns (though CM is cost-effective when hospitalisations are reduced), (3) philosophical resistance from 12-step-oriented treatment programmes.

11. ASSERTIVENESS TRAINING

11.1 Origins

Andrew Salter (1949) introduced "conditioned reflex therapy," which emphasised the expression of excitatory emotions (feelings, desires, disagreement) as the antidote to pathological inhibition. Wolpe and Lazarus adopted and formalised this into assertiveness training as a component of behaviour therapy.

11.2 The Assertiveness Spectrum

Style	Characteristics	Consequence
Passive	Fails to express needs; avoids conflict	Resentment, low self-esteem, exploitation
Aggressive	Expresses needs at others' expense; hostile	Alienation, guilt, damaged relationships
Passive-aggressive	Indirect expression of hostility	Confusion, mistrust, escalation
Assertive	Expresses needs directly, respectfully	Mutual respect, need satisfaction, self-efficacy

11.3 Techniques

1. **Psychoeducation** — distinguish assertive from aggressive and passive
2. **Modelling** — therapist demonstrates assertive responses
3. **Behavioural rehearsal (role-play)** — patient practises assertive responses in session
4. **Feedback and coaching** — therapist provides specific, constructive feedback
5. **Graded real-world practice** — start with low-stakes situations and progress
6. **Cognitive restructuring** — address beliefs that block assertiveness ("If I say no, they'll reject me")

> CLINICAL ANCHOR
>
> Assertiveness training is a core component of social skills training in schizophrenia rehabilitation, anger management programmes, and treatments for social anxiety disorder. Wolpe considered assertive behaviour to be reciprocally inhibitory to anxiety — expressing feelings and desires is physiologically incompatible with anxious avoidance.

12. EVIDENCE BASE

12.1 Evidence Summary by Disorder

Disorder	Intervention	Evidence Level	Key Findings
Specific phobias	In-vivo exposure	Very strong (Grade A)	Single-session exposure (Öst) effective in ~90% of cases; fastest treatment in all of psychotherapy
OCD	ERP	Very strong (Grade A)	60–70% of patients achieve clinically significant improvement; as effective as SRIs; combination may be superior
PTSD	Prolonged Exposure (PE)	Strong (Grade A)	Foa et al.: PE as effective as CPT and EMDR; recommended by all major guidelines
Panic disorder	Interoceptive exposure + in-vivo exposure	Strong (Grade A)	Clark (1986), Barlow (2002): exposure to feared sensations is the active ingredient
Social anxiety	Exposure + social skills training	Strong (Grade A)	Combined with cognitive restructuring in Clark & Wells model
GAD	Applied relaxation	Moderate (Grade B)	Öst: applied relaxation equivalent to CBT in some trials

Disorder	Intervention	Evidence Level	Key Findings
Depression	Behavioural Activation	Strong (Grade A)	Dimidjian et al. (2006): BA = medication > CT for severe depression
Substance use	Contingency management	Strong (Grade A)	Largest effect size of any psychosocial intervention for stimulant dependence
Insomnia	Stimulus control + sleep restriction	Strong (Grade A)	Behavioural interventions = first-line; superior to hypnotics long-term
Chronic pain	Operant behavioural programme	Moderate (Grade B)	Fordyce (1976): reinforcement of "well behaviour," extinction of pain behaviour

12.2 Behaviour Therapy vs. CBT

The relationship between behaviour therapy (BT) and cognitive behaviour therapy (CBT) is often misunderstood.

> EXAM STRATEGY
>
> On exams, distinguish clearly between BT and CBT. BT focuses on changing behaviour directly (exposure, reinforcement, skills training) without targeting cognitions. CBT adds cognitive restructuring — identifying and modifying dysfunctional thoughts and beliefs. The Jacobson et al. (1996) dismantling study and Dimidjian et al. (2006) trial suggest that the behavioural component alone may be sufficient for depression. For anxiety disorders, the core mechanism is exposure — a purely behavioural intervention — though cognitive strategies may enhance engagement and prevent relapse.

12.3 Key Behavioural Techniques Summary

Technique	Mechanism	Primary Developer	Key Indication
Systematic desensitization	Reciprocal inhibition / counterconditioning	Wolpe (1958)	Phobias (historical; largely replaced by exposure)
In-vivo exposure	Habituation / inhibitory learning	Marks (1987)	Specific phobias, agoraphobia
ERP	Exposure + response prevention	Foa & Kozak (1986)	OCD
Prolonged Exposure	Emotional processing	Foa (1986)	PTSD
Interoceptive exposure	Extinction of fear of bodily sensations	Barlow (1988)	Panic disorder
Flooding	Massive exposure / extinction	Stampfl (1967)	Phobias (rapid but high dropout)
Behavioural Activation	Increase response-contingent reinforcement	Lewinsohn (1974), Martell (2001)	Depression
Token economy	Operant conditioning / secondary reinforcement	Ayllon & Azrin (1968)	Institutional settings, schizophrenia
Contingency management	Operant reinforcement of abstinence	Higgins (1991)	Substance use disorders
Aversion therapy	Classical conditioning of aversion	Cautela (1967)	Historical; alcohol dependence
Assertiveness training	Reciprocal inhibition / skills training	Salter (1949), Wolpe (1958)	Social anxiety, interpersonal deficits

Technique	Mechanism	Primary Developer	Key Indication
Applied relaxation	Coping skill deployment	Öst (1987)	GAD, panic disorder
Biofeedback	Physiological self-regulation	Various (1960s–70s)	Headache, chronic pain, Raynaud's
Social skills training	Modelling + rehearsal + feedback	Bandura (1969)	Schizophrenia, social anxiety
Habit reversal	Awareness + competing response	Azrin & Nunn (1973)	Tics, trichotillomania

KEY INSIGHT

MNEMONIC — "BEST FACE": Key behavioural techniques for exams: **B**ehavioural Activation, **E**xposure (all forms), **S**ystematic desensitization, **T**oken economy, **F**looding, **A**ssertiveness training, **C**ontingency management, **E**RP.

End of D1: Behaviour Therapy — Deep Study

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02

Clinical Quick Reference

Behaviour Therapy — Weave Psychotherapy Vol. 5

D6: Behaviour Therapy — Quick Reference

1. Key Figures Timeline

Year	Figure	Contribution	Why It Matters
1897	Pavlov	Classical conditioning	Foundation — all learned emotional responses
1913	Watson	Behaviourist manifesto	Psychology as science of observable behaviour
1920	Watson & Rayner	Little Albert	Fear is conditioned (acquired)
1924	Mary Cover Jones	Little Peter	Fear is deconditioned — first behaviour therapy
1938	Skinner	Operant conditioning	Behaviour shaped by consequences
1949	Salter	Conditioned reflex therapy	Precursor to assertiveness training
1958	Wolpe	Systematic desensitization	Reciprocal inhibition — first systematic clinical method
1961	Bandura	Bobo doll experiment	Observational learning without direct reinforcement
1966	Ayllon & Azrin	Token economy	Operant conditioning in institutions
1967	Stampfl	Flooding/implosive therapy	Massive exposure without relaxation

Year	Figure	Contribution	Why It Matters
1974	Lewinsohn	Behavioural model of depression	Low reinforcement → depression
1977	Bandura	Self-efficacy theory	Belief in capability predicts behaviour
1986	Foa & Kozak	Emotional processing theory	How exposure works — fear activation + disconfirmation
1987	Marks	Exposure as core mechanism	Direct exposure > relaxation-based approaches
1987	Öst	Applied relaxation	Rapid relaxation as coping skill
1996	Craske	Inhibitory learning model	Exposure = new learning, not erasure
2006	Dimidjian et al.	BA for severe depression	BA = medication > cognitive therapy

2. Classical vs. Operant Conditioning

Feature	Classical Conditioning	Operant Conditioning
Pioneer	Pavlov (1897)	Skinner (1938)
Alternative name	Respondent conditioning	Instrumental conditioning
Learning mechanism	Association between stimuli (CS-US pairing)	Association between behaviour and consequence
Role of organism	Passive (responds to stimuli)	Active (operates on environment)
Response type	Involuntary/reflexive (salivation, fear, nausea)	Voluntary/skeletal (pressing lever, studying, avoidance)
Key element	Stimulus pairing (contiguity)	Reinforcement/punishment (contingency)
Extinction	CS presented without US → CR weakens	Behaviour no longer reinforced → behaviour weakens
Clinical relevance	Phobias, PTSD, panic, conditioned drug cues	Avoidance behaviour, substance use, institutional behaviour
Treatment approach	Exposure, systematic desensitization	Token economy, contingency management, BA

3. Reinforcement and Punishment Grid

Add Stimulus (+)	Remove Stimulus (−)	
Behaviour ↑	**Positive Reinforcement** — Praise after homework → more homework	**Negative Reinforcement** — Take anxiolytic → anxiety removed → more pill-taking
Behaviour ↓	**Positive Punishment** — Scold after misbehaviour → less misbehaviour	**Negative Punishment** — Remove phone after outburst → fewer outbursts

Remember: Positive/negative = add/remove. Reinforcement/punishment = increase/decrease.

4. Schedules of Reinforcement

Schedule	Pattern	Response Rate	Extinction	Clinical Example
Continuous (CRF)	Every response reinforced	Moderate	Fastest	Teaching new behaviour
Fixed Ratio (FR)	After N responses	High + pause	Fast	Token economy (5 tasks = 1 token)
Variable Ratio (VR)	After ~N responses (unpredictable)	Highest, steady	**Slowest**	Gambling; reassurance-seeking
Fixed Interval (FI)	After set time	Scallop pattern	Moderate	Medication schedules
Variable Interval (VI)	After ~set time (unpredictable)	Steady, moderate	Slow	Random check-ins

Exam rule: VR = most resistant to extinction. This is why gambling and intermittent reinforcement create the strongest habits.

5. Exposure Therapy Types

Type	Method	Indication	Key Detail
In-vivo	Real-world confrontation	Specific phobias, agoraphobia	Most effective form
Imaginal	Mental imagery of feared situation	PTSD, fears hard to recreate	Core of Prolonged Exposure (Foa)
Interoceptive	Induce feared body sensations	Panic disorder	Hyperventilation, spinning, straw-breathing
Virtual reality	Computer-simulated environment	Flying phobia, heights, PTSD	Growing evidence; practical for inaccessible stimuli
Flooding	Maximum-intensity prolonged exposure	Rapid fear reduction needed	High dropout; rarely first-line
Graded	Systematic hierarchy progression	Standard for most anxiety disorders	Patient-controlled pace
ERP	Exposure + ritual prevention	OCD	Gold standard; 60–70% response

6. Systematic Desensitization — Step-by-Step

Step	Action	Details
1	Relaxation training	PMR: 16 → 7 → 4 muscle groups; 4–6 sessions
2	Build anxiety hierarchy	10–20 items rated 0–100 SUDs; evenly spaced
3	Pair relaxation + imagery	Start at lowest; advance when anxiety = 0 on 2 trials
4	Progress up hierarchy	3–4 items per session; if SUDs > 25, return to relaxation
5	In-vivo transfer	Apply in real situations once imaginal hierarchy completed

Mechanism: Reciprocal inhibition (Wolpe) — relaxation inhibits anxiety.

Now largely replaced by: Direct exposure therapy (relaxation not essential).

7. Behavioural Activation — Core Techniques

Technique	What It Does
Activity monitoring	Hourly tracking of all activities + mood rating 0–10
Activity scheduling	Plan specific activities at specific times (action before motivation)
Mastery/Pleasure ratings	Rate each activity for accomplishment (M) and enjoyment (P)
Graded task assignment	Break overwhelming goals into small sequential steps
Values assessment	Identify what matters across life domains
TRAP → TRAC	Trigger → Response → Avoidance Pattern → replace with → Alternative Coping

Key evidence: Dimidjian et al. (2006) — BA = paroxetine > cognitive therapy for severe depression.

8. Complete Behavioural Techniques Catalogue

Technique	Mechanism	Developer	Primary Indication
Systematic desensitization	Counterconditioning	Wolpe (1958)	Phobias
In-vivo exposure	Habituation/inhibitory learning	Marks (1987)	Phobias, agoraphobia
ERP	Exposure + response prevention	Foa (1986)	OCD
Prolonged Exposure	Emotional processing	Foa (1986)	PTSD
Interoceptive exposure	Extinction of fear of sensations	Barlow (1988)	Panic disorder
Flooding	Massive extinction	Stampfl (1967)	Phobias
Behavioural Activation	↑ Response-contingent reinforcement	Lewinsohn/Martell	Depression
Token economy	Operant secondary reinforcement	Ayllon & Azrin (1968)	Institutions, schizophrenia
Contingency management	Operant reinforcement of abstinence	Higgins (1991)	Substance use
Assertiveness training	Skills training + reciprocal inhibition	Salter/Wolpe	Social anxiety, interpersonal
Applied relaxation	Rapid coping skill	Öst (1987)	GAD, panic
PMR	Muscular tension-relaxation	Jacobson (1938)	Anxiety, insomnia, pain
Biofeedback (EMG)	Physiological self-regulation	Various (1960s)	Headache, chronic pain

Technique	Mechanism	Developer	Primary Indication
Social skills training	Modelling + rehearsal + feedback	Bandura (1969)	Schizophrenia, social anxiety
Habit reversal	Awareness + competing response	Azrin & Nunn (1973)	Tics, trichotillomania
Aversion therapy	Conditioned aversion	Various	Historical; largely abandoned
Covert sensitization	Imaginal aversive pairing	Cautela (1967)	Substance use (limited evidence)
Shaping	Successive approximations	Skinner	Developmental disabilities, ABA
Participant modelling	Model demonstrates + guides patient	Bandura (1969)	Phobias
Stimulus control	Alter antecedent conditions	Bootzin (1972)	Insomnia

9. Evidence Summary by Disorder

Disorder	First-Line Behavioural Tx	Grade	NNT / Effect Size
Specific phobias	In-vivo exposure (single session, Öst)	A	~90% response rate
OCD	ERP	A	NNT ~3; d = 1.1–1.5
PTSD	Prolonged Exposure	A	d = 1.0–1.5
Panic disorder	Interoceptive + in-vivo exposure	A	70–80% panic-free
Social anxiety	Exposure + social skills	A	d = 0.8–1.2
GAD	Applied relaxation	B	Equivalent to CBT
Depression	Behavioural Activation	A	BA = medication for severe MDD
Stimulant use	Contingency management	A	d = 0.4–0.6 (largest for any psychosocial Tx)
Insomnia	Stimulus control + sleep restriction	A	Superior to hypnotics long-term
Chronic pain	Operant behavioural programme	B	↓ pain behaviour, ↑ function

10. High-Yield Exam Pearls

1. **Extinction ≠ erasure.** Extinction creates a new inhibitory memory that competes with the original fear. Explains spontaneous recovery, renewal, and reinstatement.

2. **Negative reinforcement ≠ punishment.** Negative reinforcement INCREASES behaviour by removing something aversive (e.g., avoidance reduces anxiety → avoidance increases).
3. **Variable ratio = most resistant to extinction.** Explains gambling addiction and intermittent reinforcement effects.
4. **Bobo doll:** Learning ≠ performance. All children learned; only those who expected reward performed.
5. **Self-efficacy** is the strongest predictor of whether a patient will attempt a feared task.
6. **Lazarus (1961):** Desensitization cured 72% vs. 12% with insight therapy for phobias.
7. **Dimidjian (2006):** BA = medication > cognitive therapy for severe depression.
8. **ERP for OCD:** 60–70% achieve clinically significant improvement. Response prevention is the mechanism, not a supplement.
9. **Safety behaviours** maintain fear structures. Must be eliminated during exposure.
10. **Token economies** showed that "negative symptoms" in chronic schizophrenia were partly environmental artefacts.
11. **Mary Cover Jones** (1924) = "mother of behaviour therapy" — first systematic fear elimination.
12. **Inhibitory learning model** (Craske): maximise expectancy violation, vary contexts, remove safety signals, deepen extinction.

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11. Viva Questions

Q1: What is the difference between classical and operant conditioning? Give a clinical example of each.

Classical: involuntary response learned through stimulus pairing (CS-US). Example: a patient develops nausea (CR) entering a hospital (CS) after chemotherapy (US). Operant: voluntary behaviour shaped by consequences.

Example: a patient avoids social situations (operant behaviour) because avoidance reduces anxiety (negative reinforcement).

Q2: Explain the principle of reciprocal inhibition and its clinical application.

Wolpe (1958): if a response that inhibits anxiety (e.g., relaxation) is made to occur in the presence of anxiety-evoking stimuli, the anxiety response is weakened. Application: systematic desensitization — patient is trained in PMR, then imagines feared stimuli while relaxed, progressing up a hierarchy.

Q3: Why do extinguished fears sometimes return? What are the clinical implications?

Extinction is new learning, not erasure. The original CS-US association persists; a new inhibitory association (CS → no US) competes with it. Return of fear occurs via: spontaneous recovery (passage of time), renewal (change of context), reinstatement (re-exposure to US). Implication: exposure must be conducted in multiple contexts, and patients need relapse prevention planning.

Q4: Compare flooding and systematic desensitization.

Flooding: prolonged, intense exposure to the maximum feared stimulus without relaxation. Works through extinction/habituation. Fast but distressing; high dropout. Desensitization: graded exposure paired with relaxation. Works through reciprocal inhibition/counterconditioning. Slower but better tolerated. Both effective, but direct exposure (graded or prolonged) without relaxation is now preferred.

Q5: What is Behavioural Activation and what is its evidence base for depression?

BA is a structured treatment that increases engagement in valued, rewarding activities to break the depression-withdrawal cycle (Lewinsohn, 1974; Martell, 2001). Jacobson et al. (1996) showed the behavioural component of CBT was as effective as full CBT. Dimidjian et al. (2006) showed BA was equivalent to paroxetine and superior to cognitive therapy for severe depression.

Q6: Describe the four schedules of reinforcement and their clinical significance.

FR: reinforcement after fixed number of responses — post-reinforcement pause. VR: after variable number — highest, steadiest rate; most resistant to extinction (gambling). FI: after fixed time — scallop pattern. VI: after variable time — steady, moderate rate. Clinically: VR explains persistence of gambling and reassurance-seeking; understanding schedules informs design of token economies and contingency management.

Q7: What is ERP and why is response prevention essential?

Exposure and Response Prevention for OCD: patient is exposed to obsessional triggers (contamination, harm cues) while the compulsive ritual is blocked. Without response prevention, exposure alone is counter-therapeutic — completing the ritual reinforces the belief that the ritual prevented the feared outcome. ERP has Grade A evidence; 60–70% response rate.

Q8: Explain Bandura's concept of self-efficacy and its four sources.

Self-efficacy = belief in one's capability to perform a specific behaviour. Four sources (in order of potency): (1) mastery experiences (personal success — strongest), (2) vicarious experiences (seeing similar others succeed), (3) verbal persuasion (encouragement), (4) physiological/emotional states (interpreting arousal as capability vs. incapability). Self-efficacy predicts treatment engagement and outcome across disorders.

Q9: What are the ethical concerns with aversion therapy?

(1) Historical misuse in "conversion therapy" for homosexuality — caused significant harm, now banned by professional bodies. (2) Poor long-term efficacy — conditioned aversions extinguish rapidly. (3) High dropout due to unpleasant procedures. (4) Questions about informed consent when patients are desperate or coerced. (5) Better alternatives now available (MI, CM, pharmacotherapy). Current use is extremely limited.

Q10: Compare the habituation model and inhibitory learning model of exposure therapy.

Habituation model: fear naturally decreases with prolonged, repeated exposure (within- and between-session habituation). Inhibitory learning model (Craske): exposure creates a new inhibitory memory (CS → no US) that competes with the original excitatory memory (CS → US). The inhibitory model better explains fear return and has changed clinical practice: maximise expectancy violation, vary contexts, remove safety behaviours, use deepened extinction, avoid reliance on within-session fear reduction as a progress marker.

End of D6: Behaviour Therapy — Quick Reference

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